Data Analytics (DA)

Module-1- Excel

Excel Assignment 2

DATA EXPLORATION & CLEANING REPORT

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Course : Data Analystics – Modul 1
Assignment : Excel Assignment -2 Data Exploration
Submission Date : 07-11-2026

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1. Introduction and Objective

The aim of this assignment is to clean and prepare the Product Dataset using Microsoft Excel. The work includes missing values, text corrections, duplicate removal, column transformation and formatting.

2. Dataset Overview

Item	Result
Source worksheet	RAW DATA
Original records	34
Original columns	6 — Product ID, Product Name, Brand Name, Price ($), Quantity, Category
Exact duplicate copies removed	3
Unique records after duplicate removal	31
Missing Price values	3
Missing Category values	4
Category typo	Electroni → Electronics

The counts above are based on the uploaded workbook's RAW DATA sheet. Duplicate detection is performed across all six original fields, not by Product ID alone.

3. Step-by-Step Data Cleaning

1. Missing Value Handling

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronic
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronic
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni

Task 1: Check missing values in the Price column

Step 1 – Open your Excel file

Open RAW DATA.xlsx.

Step 2 – Find blank prices

Select the Price column (Column D).

Go to:

1. **Home**
2. **Find & Select**
3. **Go To Special**
4. **Select Blanks**
5. **Click OK**

Example Excel formula to count missing prices:

```
=COUNTBLANK(Data_set4[@Category])
```

. Missing Value Handling

Explanation: Identified missing values in the dataset. Found 3 missing prices and 4 missing categories. Missing categories were suggested based on product names, while unknown prices were kept blank for verification.

Screenshot:



Expected result: 3

Task 2: Handle missing Category values

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronic
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronic
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni

Step 1 – Select the Category column

Select Column F.

Step 2 – Find blank cells

Use the same Go To Special → Blanks method.

Step 3 – Check the product name and brand

Use product knowledge to identify the appropriate category.

Step 4 – Fill the missing categories

Enter the suggested category only after verification.

Excel formula to count missing categories:

=IF(ISBLANK(Data_set4[@Category]),"unknown",Data_set4[@Category])

Screenshot:

```
=COUNTBLANK(Data_set4[@Category])
```

```
=IF(ISBLANK(Data_set4[@Category]),"unknown",Data_set4[@Category])
```

Expected result: 4

Raw Data

						Missing Price Values		Missing Category values	
						IFBLANK	Replace	COUNTBLANK	Replace
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price ($)	Price	Category2	Category2
28-JAN-US	laptop	Dell	1000	30	Electronicss	FALSE	0	0	Electronicss
15-FEB-US	Sneakers	Nike	80	15	Fashion	FALSE	0	0	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	FALSE	0	0	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronicss	FALSE	0	0	Electronicss
22-MAY-US	Backpack	North Face	70	20		FALSE	0	1	unknown
07-JUN-UK	Headphones	Sony		45	Electroni	TRUE	unknown	0	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	FALSE	0	0	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen	FALSE	0	0	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronicss	FALSE	0	0	Electronicss
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	FALSE	0	0	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronicss	FALSE	0	0	Electronicss
25-NOV-AU	Sneakers	Adidas	90	40		FALSE	0	1	unknown
08-DEC-DE	Coffee Maker	Nespresso	120	35		FALSE	0	1	unknown
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni	FALSE	0	0	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni	FALSE	0	0	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	FALSE	0	0	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni	FALSE	0	0	Electroni
27-JAN-IT	laptop	Asus	980	10	Electronicss	FALSE	0	0	Electronicss
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	FALSE	0	0	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor	TRUE	unknown	0	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronicss	FALSE	0	0	Electronicss
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	FALSE	0	0	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30		FALSE	0	1	unknown
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	FALSE	0	0	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronicss	FALSE	0	0	Electronicss
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	TRUE	unknown	0	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	FALSE	0	0	Kitchen
16-APR-ES	headphones	Bose	250	20	Electroni	FALSE	0	0	Electroni
07-MAR-CA	Dress	Zara	60	30	Fashion	FALSE	0	0	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	FALSE	0	0	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronicss	FALSE	0	0	Electronicss
02-DEC-CA	Jeans	Levi's	50	50	Fashion	FALSE	0	0	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronicss	FALSE	0	0	Electronicss
09-JUL-FR	Watch	Casio	100	20	Accessories	FALSE	0	0	Accessories

2. Find and Replace Corrections

Explanation: Used Excel's Find and Replace feature to correct spelling mistakes and maintain consistency in category names**.**

Example:

- **Electroni → Electronics**

- **Explanation:** Standardising spelling and capitalisation prevents the same product or category from being treated as different values in filters, PivotTables and charts.

```
=PROPER(Data_set411[@[Product Name]])
```

Find and Replace	
Spelling correction	Use Proper Formula
Product Name	Product Name
Electronics	Laptop
Fashion	Sneakers
Kitchen	Coffee Maker
Electronics	Smartphone
	Backpack
Electronics	Headphones
Fashion	T-Shirt
Kitchen	Blender
Electronics	Tablet
Outdoor	Hiking Boots
Electronics	Laptop
	Sneakers
	Coffee Maker
Electronics	Smartwatch
Electronics	Headphones
Accessories	Laptop Bag
Electronics	Smartwatch
Electronics	Laptop
Fashion	Sunglasses
Outdoor	Camping Tent
Electronics	Camera
Kitchen	Microwave
	Fitness Tracker
Accessories	Laptop Bag
Electronics	Smartphone
Fashion	Sunglasses
Kitchen	Blender
Electronics	Headphones
Fashion	Dress
Kitchen	Toaster
Electronics	Fitness Tracker
Fashion	Jeans
Electronics	Laptop
Accessories	Watch

3. Duplicate Removal

Selected all six dataset columns and used Data → Remove Duplicates. Removed 3 exact duplicate copies. The dataset now contains 31 unique records.

Excel steps: (1) Select the full table including all six fields.

(2) Open Data → Remove Duplicates.

(3) Tick all six columns.

(4) Click OK.

(5) Record the number removed and confirm the remaining row count.



RAW DATA

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
27-JAN-IT	laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	headphones	Bose	250	20	Electroni
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

remove them if any.

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
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03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
09-JUL-FR	Watch	Casio	100	20	Accessories

Explanation:

Removing exact duplicate rows prevents double counting and improves the reliability of summaries. Do not remove records merely because one field, such as Product Name, repeats.

4. Column splitting and merging

Column Splitting

Explanation:

Used Text to Columns with a hyphen (-) to split Product ID into date token and Country Code. Merged Brand Name and Product Name using =TRIM(C8&" "&B8). The Product ID does not contain a year.

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronic
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronic
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni

Split: First Name and Last Name

Excel steps for splitting: (1) Select Product ID.

(2) Choose Data → Text to Columns.

(3) Select Delimited → Next.

(4) Tick Other and type a hyphen -.

(5) Finish and rename the resulting fields appropriately. Keep the date token as text until the year is confirmed.

SPLIT

" Manufacturing Date	"Country Code".
28	US
15	US
03	US
11	US

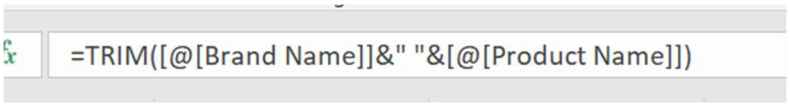
Column Merging

To merge Brand Name and Product Name, create Product Brand and enter =TRIM(C8&" "&B8), then fill down. Copy and Paste Special → Values if a fixed result is required

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronic
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronic
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni

Merge: Brand Name and Product Name

Use Formula



MERGE
"Product Brand".
Dell laptop
Nike Sneakers
Keurig Coffee Maker
Samsung smartphone
North Face Backpack
Sony Headphones
Adidas T-shirt

Number and Date Formatting

Formatted the Price column as Currency using the dollar symbol shown in the source. The date format DD-MM-YYYY can be applied.

RAW DATA

Product ID	Price ($)
28-Jan-26	1000
15-Feb-26	80
03-Mar-26	130
11-Apr-26	900
22-May-26	70
07-Jun-26	
19-Jul-26	30
23-Aug-26	90
05-Sep-26	500
14-Oct-26	130

Format Price as Currency: select the Price values → Home → Number group → Currency (or Ctrl+Shift+4 where the local Excel settings support it). Choose the required currency symbol and decimal places. The source column is labelled Price ($), so retain the dollar symbol unless the instructor specifies otherwise.

Currency Format	"DD-MM-YYYY " Format
"Price"	"Manufacturing Date"
₹ 1,000.00	28 January 2026
₹ 80.00	15 February 2026
₹ 130.00	03 March 2026
₹ 900.00	11 April 2026
₹ 70.00	22 May 2026

Format Manufacturing Date as DD-MM-YYYY only after a valid year has been verified and a real Excel date value has been created. Then select the date cells → Ctrl+1 → Number → Custom → type dd-mm-yyyy. Formatting a text token such as 28-JAN as a date does not create the missing year.

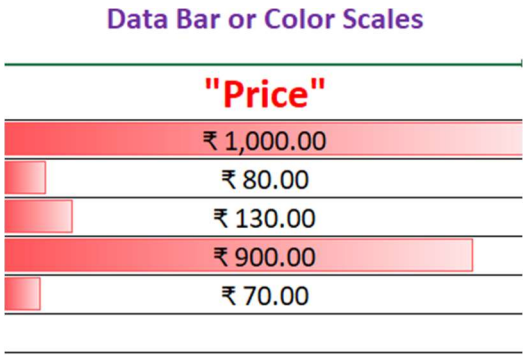
5. Conditional formatting results

Explanation: Applied Conditional Formatting to highlight missing values, duplicate records and important price information using different colours**.**

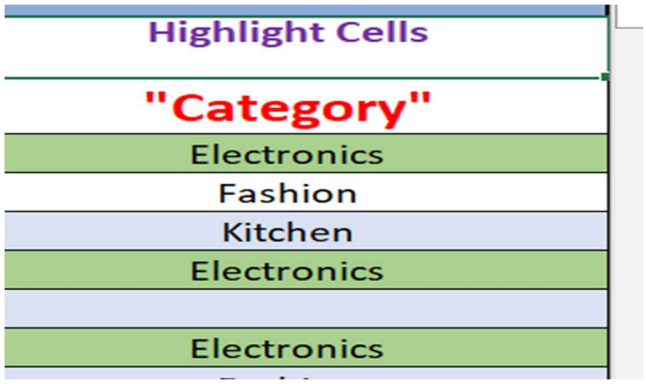
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronic
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronic
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni

Price

For Price, select the numeric Price range and choose Home → Conditional Formatting → Data Bars (or Color Scales).



For Category, select the category range → Conditional Formatting → New Rule → Format only cells that contain → Specific Text → containing → Electronics; choose a visible fill colour and apply.



Explanation: Data bars or colour scales provide a quick visual comparison of price values. The Electronics rule highlights all matching category cells and makes the category easy to identify.

Final Validation Checklist

Requirement	Validation / status
Missing Price values	3 identified; keep blank and verify from supplier source
Missing Category values	4 identified; proposed category mapping documented for verification
Text standardisation	Product Name title case; Electroni corrected to Electronics
Duplicate removal	3 exact duplicate copies removed; 31 unique records remain
Column splitting	Product ID split using hyphen; year not present in source ID
Column merging	Product Brand = Brand Name + Product Name
Price formatting	Apply Currency format to numeric Price values
Date formatting	Apply DD-MM-YYYY only after year confirmation
Conditional formatting	Price data bars/colour scale and Electronics category rule

The dataset was checked for missing values, inconsistent text and duplicate records. Three duplicate copies were removed, leaving 31 unique records. The remaining cleaning and formatting steps are documented above

Conclusion

Excel cleaning tools help improve the accuracy, consistency and readability of data. Submit the cleaned workbook together with this report.